ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C1 · MATERIALS

Sorting materials

★☆☆
Grade 3

Learning goals

- Use property words.
- Sort materials.

1 Write the material each object is made of. [4]



metal



glass



wood



fabric (cotton)

2 Sort these materials. [6]

SORT THESE

steel stone cotton wool wool glass sponge

Hard	Soft
stone	cotton wool
steel	sponge
glass	wool

3 Will a magnet attract it? [5]

Object	Material	Magnetic?
nail	iron	yes
drink can	aluminium	no
paper clip	steel	yes
ruler	plastic	no
coin	brass or steel	test it! some are, some are not

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C1 · MATERIALS

The right material for the job

★★☆
Grade 3

Learning goals

- Explain choices of materials.

1 Why is a window made of glass, not wood? [2]

Glass is transparent so light can come in; wood is opaque.

2 Why would a paper umbrella be a bad idea? [2]

Paper is absorbent and weak; it would get soggy and tear in the rain.

3 Odd one out. Circle it and say why. [2]

	A	B	C	D
group	iron nail	steel pin	aluminium can	paper clip

C, aluminium can: it is the only one that is not magnetic.

4 Design a raincoat for a school trip. Label the materials and say why you chose them. [3]

My raincoat

e.g. plastic coat (waterproof), cotton lining (soft), shiny strips (easy to see).

ANSWER KEY

INVESTIGATE · C1 · MATERIALS

Which material soaks up the most water?

★★☆ Grade 3

Learning goals

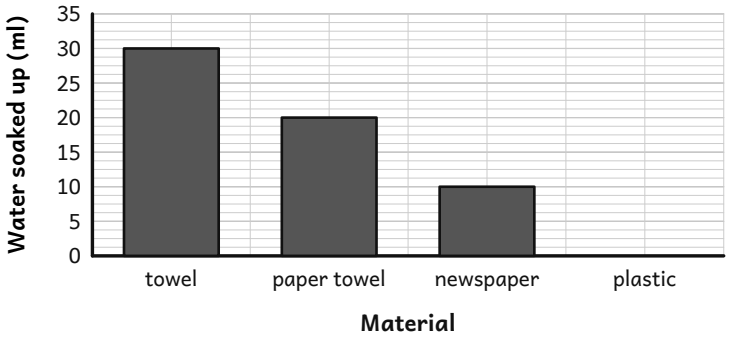
- Do a fair test.
- Draw a bar chart and conclude.

Class 3A put a 10 cm square of each material into a dish with 50 ml of water. After one minute they lifted it out and measured how much water was left in the dish.

1 **Complete the table: how much water did each material soak up?** [4]

Material	Water left (ml)	Water soaked up (ml)
cotton towel	20	30
paper towel	30	20
newspaper	40	10
plastic	50	0

2 **Draw a bar chart of the water soaked up.** [3]



3 **How did they make it a fair test?** [2]

Same size of material, same amount of water, same time.

4 **Which material would you use to mop up a spill? Why?** [1]

the cotton towel: it is the most absorbent

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C1 · MATERIALS

Jute: the golden fibre

★★★
Grade 3

Learning goals

- See how properties make a material useful.

Science | Farming | Environment

Jute is a tall plant that grows well in the warm, wet fields of Bangladesh. Farmers soak the stems in water, then peel off long, shiny, golden threads called **fibres**. Because of its colour and value, jute is called the **golden fibre**.

Jute fibres are **strong** and **flexible**, so they are made into rope, sacks, mats and bags. Unlike plastic, jute **rots away** in the soil when it is thrown out. Bangladeshi scientists have also made a thin, plastic-like bag from jute, so that fewer plastic bags end up in rivers and drains.

Based on information about jute farming and jute products in Bangladesh.

1 **Write two properties of jute fibre.** [2]

strong, flexible (also shiny, golden)

2 **Why is a jute bag better for the environment than a plastic bag?** [2]

It rots away in the soil; plastic stays for hundreds of years and blocks drains.

3 **Name two things made from jute.** [2]

rope, sacks, bags, mats, carpets

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C1 · MATERIALS

Show what you know

★★★
Grade 3

Learning goals

- Use material words.

1 Find the words.

J	U	T	E	C	R	M	A	Y
S	H	O	O	Y	N	N	W	L
J	B	V	G	M	M	F	O	O
Y	S	E	W	L	E	I	O	I
R	N	H	O	R	A	T	D	N
E	G	H	I	S	U	S	A	D
U	T	N	A	N	O	I	S	L
U	T	C	J	R	Y	F	D	K
B	M	H	R	J	D	E	T	A

glass hard jute metal shiny soft wood

2 Which material is transparent? [1]

○ wood
● **glass**
○ brick

3 Which material is magnetic? [1]

○ plastic
○ aluminium
● **iron**

4 Challenge. A builder needs a material for a bridge. Which properties must it have? [2]

strong, stiff, waterproof, does not rot or rust easily